

Dominik Schrempf

Klosterneuburg, Austria

* December 6, 1986

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🌐 blog.composef.org/about

Professional experience

- May 2025 – present **Haskell software consultant**
Well-Typed LLP, London, UK
- Project Development of `hs-bindgen`, a tool for automatic generation of Haskell bindings from C header files
Haskell, Compiler development, C foreign function interface
- Jun 2023 – April 2025 **Software consultant**
TNG Technology Consulting, Munich, Germany
- Project Analysis and classification of neurobiological time-series data using a large variety of methods including state of the machine learning methods and deep neural networks
Python, NumPy/SciPy stack, scikit-learn, Aeon, Tensorflow
- Project Full stack development of a business process orchestrator
TypeScript, AWS with infrastructure as code, PostgreSQL, React
- Jan 2018 – May 2023 **Postdoc in computational biology**
Eötvös Loránd University, Budapest, Hungary
- Research Probabilistic models for estimating ages of ancestral species
Bayesian inference with Markov chain Monte Carlo algorithms; Maximum likelihood inference with expectation maximization and gradient ascent; Computational simulations with Haskell
- Mar 2017 – Aug 2017 Research fellow
University of St. Andrews, United Kingdom

Education and training

- Mar 2016 – May 2016 Visiting fellow
Aarhus University, Denmark
Advisor: Asger Hobolth
- Sep 2013 – Aug 2017 **Doctor of Philosophy in computational biology**
University of Veterinary Medicine Vienna, Austria
Research: Inference of evolutionary trees using DNA sequences
- Oct 2010 – Apr 2013 **Master of Science in technical physics**
Vienna University of Technology, Austria
Research: Statistics of ion-induced electron emission
- Feb 2010 – Sep 2010 Visiting student
Pontificia Universidad Católica de Chile, Santiago de Chile
- Oct 2006 – Feb 2010 **Bachelor of Science in technical physics**
Vienna University of Technology, Austria
Research: Interacting hydrogen atoms and their application as qubits

1997 – Jul 2005 **High school degree**
Scientific High School, Brucknerstraße Wels, Austria

Awards

- 2018 [Promotio sub auspiciis praesidentis rei publicae](#): Highest honor for academic success of Austria; Austrian Award of Excellence (~ 10 000€)
- 2016 Residence grant, Aarhus University, Denmark (~ 3000€)
- 2010 Joint Study Award for an exchange semester in Santiago de Chile
- 2007 – 2009, 2012 Academic scholarship, Vienna University of Technology (~ 900€ yearly)

Languages

German	Primary fluency	<i>Mother tongue</i>
English	Primary fluency	<i>Working language for more than 10 years</i>
Spanish	Professional working proficiency	
Hungarian	Limited working proficiency	<i>Partly spoken in my family</i>

Computational skills

Programming languages	██████	Haskell	<i>Development of libraries and applications</i>
	██████	Nix	<i>Software packaging, system deployment</i>
	██████	TypeScript	<i>Full stack application development</i>
	██████	Python	<i>Data science, machine learning and plotting</i>
	██████	C, C++	
	██████	Rust	
	██████	Fortran	
Databases	██████	PostgreSQL	
	██████	GraphQL	
Mathematics and statistics	██████	R	
	██████	Mathematica	
	██████	Matlab	
Tools	██████	Emacs	<i>Everyday use</i>
	██████	Bash, Linux core utilities	
	██████	LaTeX	<i>Scientific publications and presentations</i>
	██████	(Ma)git	
	██████	LibreOffice	
Cloud services	██████	Amazon Web Services	
	██████	GitHub Actions	
Operating system	██████	Linux	<i>NixOS; previously Arch Linux and Debian</i>

Maintained packages on the Haskell package archive

mcmc	<i>Markov chain Monte Carlo sampler with advanced algorithms</i>
ELynx Suite	<i>Library and tool set for computational biology</i>
circular	<i>Circular fixed-size mutable stacks</i>
covariance	<i>Estimation of large-dimensional covariance matrices</i>
dirichlet	<i>Multivariate Dirichlet distribution</i>
pava	<i>Computation of greatest convex majorants and least concave minorants</i>

Teaching experience

- Supervised students Lénárd Szánthó (2020 – 2022), Lukas Weilguny (2016)
- 2014 – 2015 University of Veterinary Medicine Vienna (Austria)
Supervised courses: Bioinformatics in Biomedicine, Next Generation Sequencing
- 2008 – 2012 Vienna University of Technology (Austria)
Supervised courses: Basic Principles of Physics 1–3, Modelling in Physics, Quantum Mechanics 2 and Laboratory Work
- 2010 – 2011 Tutor, Lernquadrat (Tutored pupils aged 11 – 18 in Mathematics and Physics), Vienna (Austria)

Official functions

- 2014 – 2017 Member of the Representatives of the PhD students of the University of Veterinary Medicine Vienna
- 2014 – 2017 Member of the Curricular Commission of the PhD program of the University of Veterinary Medicine Vienna
- 2014 – 2015 Student speaker of the Vienna Graduate School of Population Genetics

Scientific publications

I have worked on 24 scientific papers, 19 of which have already been published in renowned scientific journals ([Google Scholar profile](#)).

Extracurricular activities

- Oct 2005 – Sep 2006 Paramedic (civil service), *Red Cross*, Grieskirchen, Austria
- In my free time I enjoy spending time with my family, singing in a choir and making music with the saxophone and the piano. I am fond of juggling and winter sports as well as hiking in the mountains with friends. I try to maximize bike usage and reduce my carbon footprint. I like to travel by train and to get to know new cultures and ways of life.